

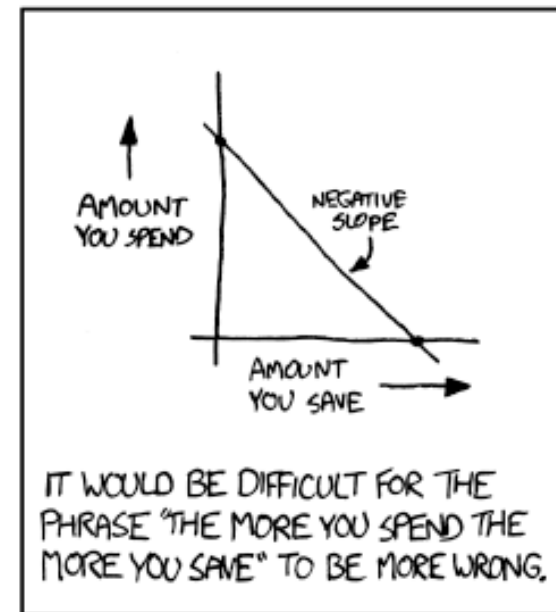
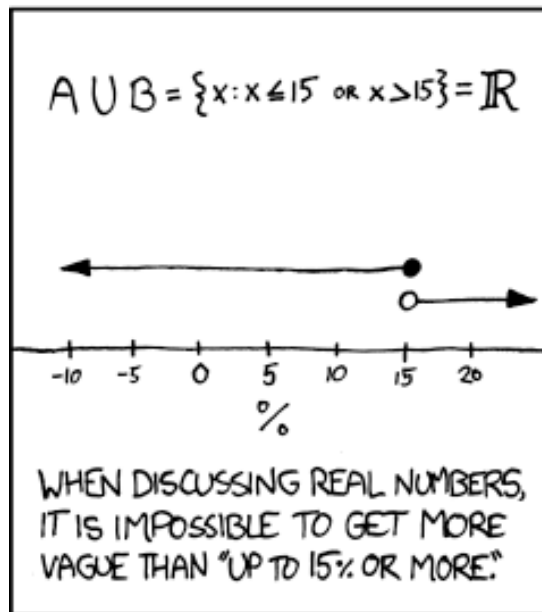
CS 237: PROBABILITY IN COMPUTING

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More on Expectation: Linearity and LOTUS

MATHEMATICALLY ANNOYING ADVERTISING:



<https://xkcd.com/870/>

RECAP OF EXPECTED VALUE

- ▶ **Definition:** Let X be a discrete random variable. The expected value (or expectation or mean) of X is

$$\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}[\{\omega\}]$$

- ▶ Let X be a random variable with PMF f (discrete) or PDF f (continuous)

discrete: $\mathbb{E}(X) = \sum_{x \in \mathbb{R}} x \cdot f(x)$

continuous: $\mathbb{E}(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$

RECAP: DISCRETE UNIFORM

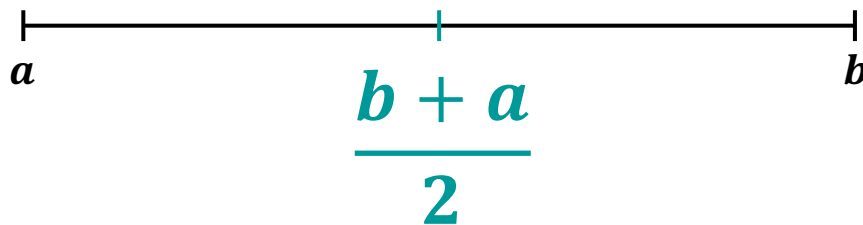
- ▶ Let X be a value chosen uniformly at random from the set $\{x_1, x_2, \dots, x_n\}$
- ▶ Then $\mathbb{E}(X) = \frac{1}{n} \sum_{i=1}^n x_i$

RECAP: CONTINUOUS UNIFORM

- ▶ Suppose X is chosen uniformly at random from the interval $[a, b]$. The PDF is $f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$
- ▶
$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{\infty} x \cdot f_X(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left. \frac{x^2}{2} \right|_a^b = \frac{1}{b-a} \frac{b^2 - a^2}{2} = \frac{b+a}{2}\end{aligned}$$

RECAP: CONTINUOUS UNIFORM

- ▶ Suppose X is chosen uniformly at random from the interval $[a, b]$. The PDF is $f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$
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$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{\infty} x \cdot f_X(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left. \frac{x^2}{2} \right|_a^b = \frac{1}{b-a} \frac{b^2 - a^2}{2} = \frac{b+a}{2}\end{aligned}$$
- ▶ **Note:** $\mathbb{E}(X)$ is the midpoint of the interval $[a, b]$



ASK THE AUDIENCE

- ▶ **Experiment:** toss a fair coin until we obtain Heads
- ▶ Let X be the number of tosses

The expectation of X is

(A) 1.5 (B) 2 (C) 4 (D) infinite

SUM OF GEOMETRIC SERIES

- ▶ Find $\sum_{k=1}^{\infty} \frac{1}{2^k} =$
- ▶ Find $\sum_{k=0}^{\infty} ar^k =$

SOLUTION

You toss a fair coin until you get heads.

Let X be the number of tosses. What is $\mathbb{E}(X)$?

$$\mathbb{E}(X) = \sum_{a=1}^{\infty} a \cdot \mathbb{P}(X = a) =$$

$$\Pr(X = a) = \mathbb{P}(\underbrace{\{T \dots T\}}_a H) =$$

$$a = 1: \frac{1}{2}$$

$$a = 2: \frac{1}{2^2} + \frac{1}{2^2}$$

$$a = 3: \frac{1}{2^3} + \frac{1}{2^3} + \frac{1}{2^3}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

ASK THE AUDIENCE

- ▶ **Experiment:** toss a fair coin until we obtain Heads
- ▶ Let X be the number of tosses. Another way to compute it...

LINEARITY OF EXPECTATION

- ▶ **Theorem:** For **any** random variables X and Y

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

- ▶ The theorem holds even if X and Y are dependent!



LINEARITY OF EXPECTATION

- ▶ **Theorem:** For **any** random variables X and Y

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

- ▶ The theorem holds even if X and Y are dependent!
- ▶ More generally:

$$\mathbb{E}(aX + b) = a \cdot \mathbb{E}(X) + b \quad \text{for any constants } a, b \text{ and r.v. } X$$

$$\mathbb{E}(c_1X_1 + c_2X_2 + \cdots + c_nX_n) = c_1\mathbb{E}(X_1) + c_2\mathbb{E}(X_2) + \cdots + c_n\mathbb{E}(X_n)$$

for any **constants** $c_1, \dots, c_n$ and random variables $X_1, \dots, X_n$



SUMS OF RANDOM VARIABLES

- ▶ **Experiment:** roll two fair dice
- ▶ Let X be the number on the 1st die and Y be the number on the 2nd die
- ▶ Let S be the sum of the two rolls, i.e., $S = X + Y$
- ▶ If X and Y are independent, we can find $\mathbb{E}[S]$ using the definition and some casework. If they are not independent, it's not as clear what to do. But using linearity of expectation:

INDICATOR RANDOM VARIABLES

- ▶ An **indicator random variable** takes on two values: 0 and 1. Typically, it indicates whether some event occurred or not.
- ▶ **Lemma.** For an indicator random variable Y ,
$$\mathbb{E}(Y) = \mathbb{P}(Y = 1).$$

Proof:

INDICATOR RANDOM VARIABLES

- ▶ An **indicator random variable** takes on two values: 0 and 1. Typically, it indicates whether some event occurred or not.

- ▶ Lemma. For

An **indicator random variable** Y
for an event E is

$$Y = \begin{cases} 1 & \text{if } E \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Proof:

Method for computing expectation of a
random variable X :

Represent X as a sum of indicators and use
linearity of expectation.

ASK THE AUDIENCE

You have a coin that shows heads with probability $3/4$. Let X be the number of heads in 1000 tosses of your coin. You represent X as the sum: $X = X_1 + X_2 + \cdots + X_{1000}$. What is X_1 ?

- A. $3/4$.
- B. The number of heads.
- C. The probability of heads in toss 1.
- D. The number of heads in toss 1.
- E. None of the above.

ASK THE AUDIENCE

You have a coin that shows HEADS with probability $3/4$. Let X be the number of HEADS in 1000 tosses of your coin.

What is the expectation of X ?

- A. $3/4$.
- B. $4/3$.
- C. 500.
- D. 750.
- E. None of the above.

RANDOM PROOFS

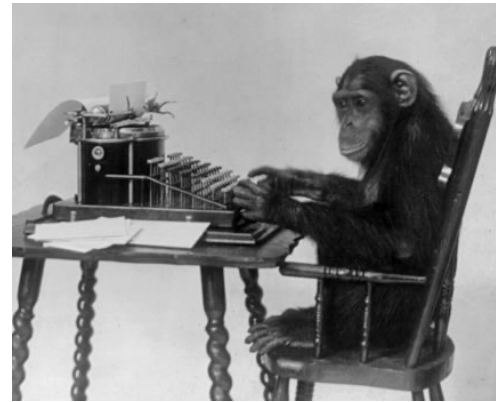
- ▶ Infinite monkey theorem:



Given enough time, a hypothetical chimpanzee typing at random would, as part of its output, almost surely produce one of Shakespeare's plays (or any other text).

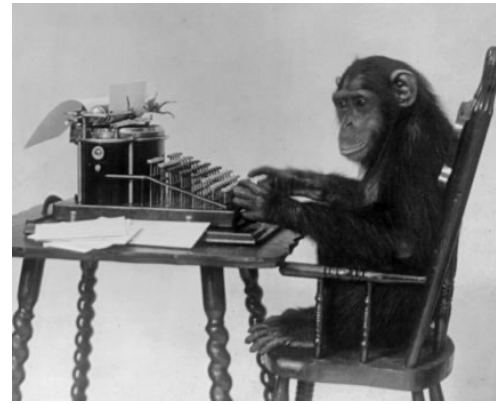
https://en.wikipedia.org/wiki/Infinite_monkey_theorem

RANDOM PROOFS



- ▶ Suppose that the monkey types on a 26-letter keyboard
- ▶ Each letter is chosen independently and uniformly at random from the 26 letters
- ▶ Suppose that the monkey types $n = 1,000,000$ letters
- ▶ Let X = number of times the sequence "proof" appears
- ▶ What is $\mathbb{E}(X)$?

RANDOM PROOFS



ASK THE AUDIENCE

3 students exchange their hats using a random permutation, so that everybody gets a random hat. Let X be the number of students that got their own hats. What's the expectation of X ?

A. $3/4$

B. 1

C. $3/2$

D. None of the above.

ASK THE AUDIENCE

If students 1,2,3 got hats 2,1,3 then $X=1$.

- ▶ PDF of X :

$$\mathbb{P}(X = 0) = \frac{1}{3}, \mathbb{P}(X = 1) = \frac{1}{2}, \mathbb{P}(X = 3) = \frac{1}{6}.$$

EXAMPLE: RANDOM HATS

- ▶ Permutations
 - ▶ n students exchange their hats, so that everybody gets a random hat
 - ▶ Let X be the number of students that got their own hats.
- ▶ What's the expectation of X for general n ?

LAW OF THE UNCONSCIOUS STATISTICIAN (LOTUS)

- ▶ We saw earlier several examples of random variables that are functions of another random variable
- ▶ Examples: $Y = 3X + 1$ $Y = X^2$ $Y = |X|$
- ▶ So how about computing $\mathbb{E}[g(X)]$, say for discrete X ? $g(X)$ is a random variable, so:

$$\mathbb{E}[g(X)] = \sum_{x \in \mathbb{R}} x \cdot \mathbb{P}[g(X) = x]$$

But what if we want to use the PMF for X ?

LAW OF THE UNCONSCIOUS STATISTICIAN (LOTUS)

- Law of the unconscious statistician: if X is a random variable then for any function $g: \mathbb{R} \rightarrow \mathbb{R}$,

X is discrete:
$$\mathbb{E}[g(X)] = \sum_{x \in \mathbb{R}} g(x) \cdot \mathbb{P}[X = x]$$

Not the same formula! But LOTUS says they are equal.

$$\mathbb{E}[g(X)] = \sum_{x \in \mathbb{R}} x \cdot \mathbb{P}[g(X) = x]$$

X is continuous:
$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx$$

FUNCTIONS OF RANDOM VARIABLES

- ▶ Let X be a random variable that is uniform over $[0, 1]$

$$f_X(x) = \begin{cases} 1 & \text{if } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f_X(x) dx = \int_0^1 x^2 dx = \frac{x^3}{3} \Big|_0^1 = \frac{1}{3}$$

LAW OF THE UNCONSCIOUS STATISTICIAN (LOTUS)

- ▶ Let X be uniform over $\{-4, \dots, 4\}$. Then PMF of X is

$$f_X(x) = \begin{cases} \frac{1}{9} & \text{if } x \in \{-4, \dots, 4\} \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}(X^2) =$$